

National Real-Time Payments System – Architecture Report

Executive Summary

The National Real-Time Payments (RTP) platform enables instant digital financial transactions across participating banks and payment providers.

The architecture prioritizes scalability, security, and resilience while supporting continuous operation and high transaction throughput.

1. Introduction

The financial services industry is evolving rapidly with growing demand for instant payment capabilities.

Traditional batch-based systems struggle to meet modern customer expectations and regulatory initiatives promoting faster payments.

2. Business Drivers

Growth in digital banking adoption and mobile payments.

Need for real-time financial transactions.

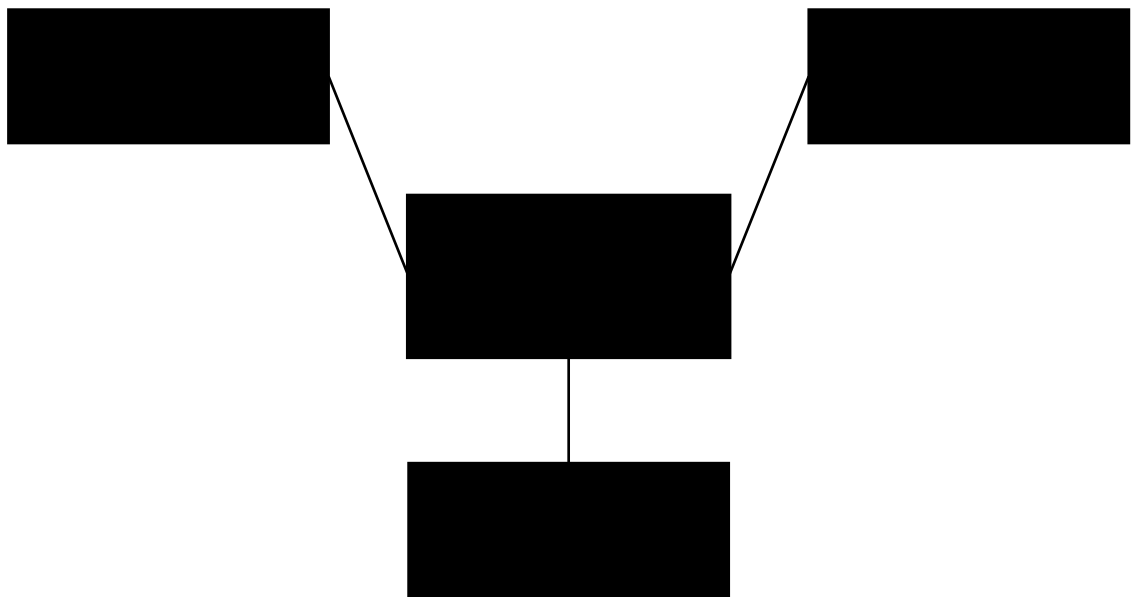
Increasing transaction volumes and regulatory expectations.

3. Architecture Overview

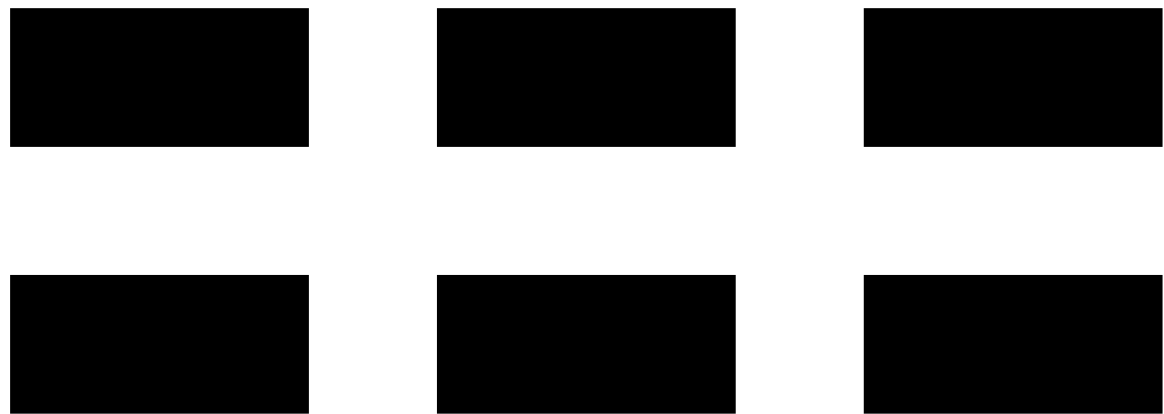
The system follows a microservices-based architecture with event-driven communication patterns.

Services are independently deployable and horizontally scalable.

C4 Context Diagram



C4 Container Diagram



4. Core Services

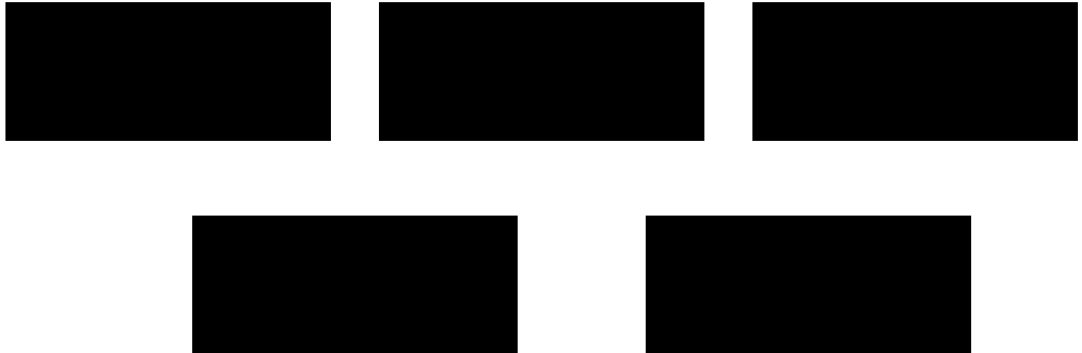
API Gateway manages secure access.

Payment Service handles transaction execution.

Fraud Detection analyzes transaction risk.

Settlement Service updates account balances.

C4 Component Diagram



5. Transaction Flow

A payment request enters through the API gateway.

The system validates and evaluates fraud risk before executing settlement.

6. Scalability Strategy

Microservices allow horizontal scaling.

Container orchestration platforms enable dynamic resource allocation.

7. Security and Compliance

End-to-end encryption protects financial data.

Role-based access control ensures secure system access.

8. Implementation Roadmap

Phase 1 establishes core infrastructure and payment processing.

Phase 2 integrates fraud detection and settlement services.

Phase 3 focuses on optimization and scaling.

9. Benefits

Instant payments improve customer experience.

Scalable architecture supports future transaction growth.

Conclusion

The proposed architecture provides a resilient and scalable foundation for national real-time payments.

It supports financial innovation and strengthens digital payment ecosystems.